

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 26)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1	.	1
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Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0	1	.	2
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Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

[illegible]

0	1	.	3
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Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

1

2

3

10

0 2 . 1 Explain why **proxy data** are needed to study the early conditions on Earth.
[1 mark]

0 2 . 2 Complete Table 1 to show what each analytical technique can be used to estimate.
[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

0 2 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

0 2 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

10

0 3 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

0 3 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

03 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

.....

.....

5

04 . 1

Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

.....

.....

.....

04 . 2

Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

1

.....

2

.....

3

.....

4

.....

04 . 3

Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale

.....

.....

.....

Large blue butterfly

.....

.....

.....

10

0 5 . 1

Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

.....

.....

.....

Soil maintenance

.....

.....

.....

Interspecies relationships

.....

.....

.....

0 5 . 2

Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

.....

.....

0 5 . 3

Explain why elephants are particularly important seed dispersers.

[2 marks]

10

0 6

Table 1 shows features of some fishing methods.

0 6 . 1

Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage
Demersal trawling		
Purse seining		
Pelagic long-lining		

06 . 2

Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

06 . 3

Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

10

07 . 1

Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

07 . 2

Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

07 . 3

Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

07 . 4

Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

10

08

A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

08 . 1

Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

08 . 2

Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

08 . 3

The test gave $p = 0.02$. Explain what this means.

[2 marks]

08 . 4

Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

08 . 5

Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

10

0	9	.	1
---	---	---	---

Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0	9	.	2
---	---	---	---

Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

10

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

ESP/7447/2/SET26

Extra space

10 . 2

State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

OR

1	1	.	2
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Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS